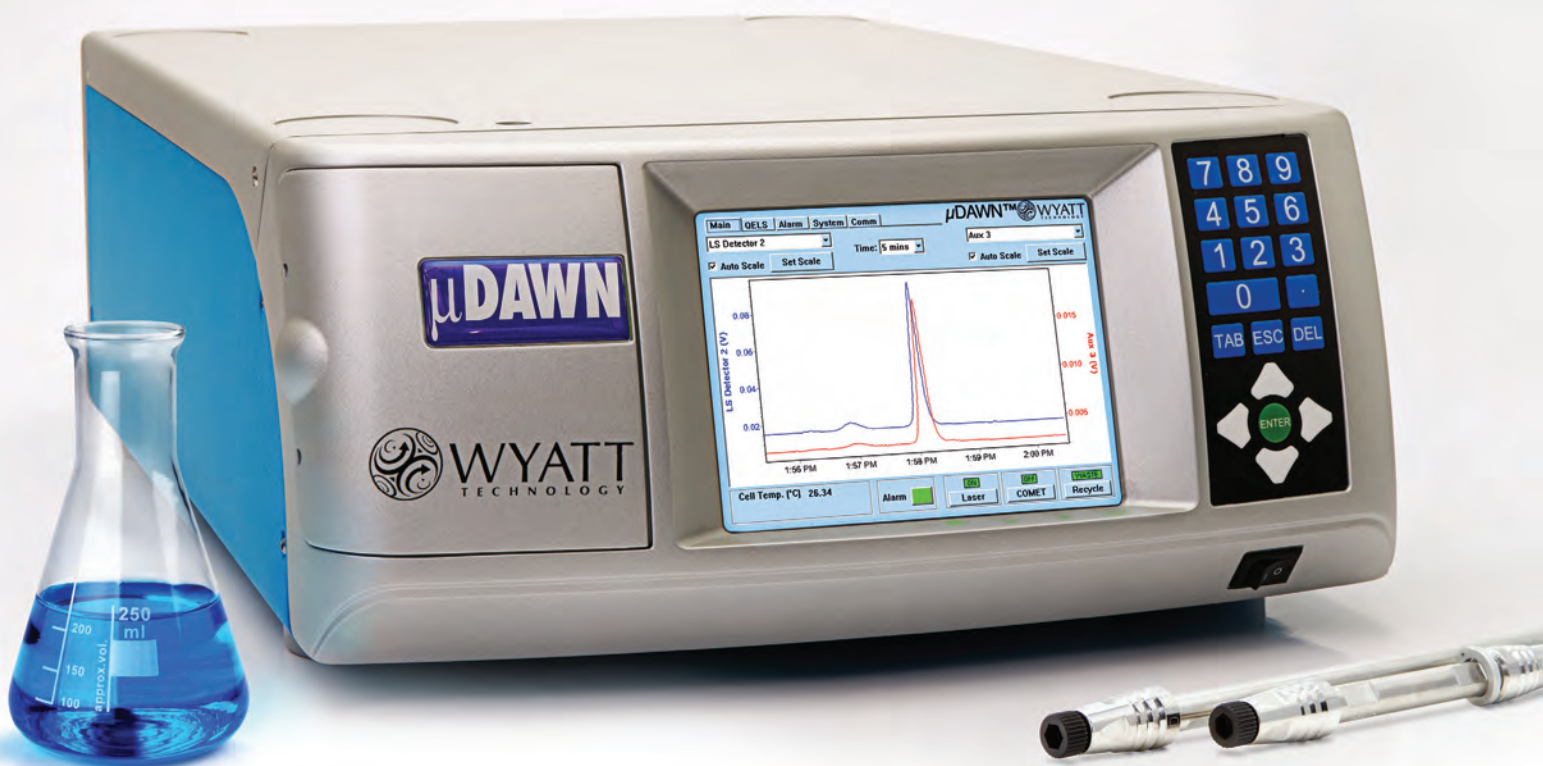




μ DAWN™

The world's first multi-angle
light scattering detector for UHPLC



μDAWN™

Absolute molar mass and size for UHPLC

The μDAWN brings all the benefits of multi-angle light scattering (MALS) analysis to ultra-high-performance liquid chromatography (UHPLC). Combine a μDAWN with any UHPLC system, a size exclusion column and the Optilab® UT-rEX™ refractive index detector for a complete μSEC-MALS system to characterize proteins and polymers for:

- ✓ Solution properties of macromolecules
- ✓ Molar mass, independent of elution time
- ✓ Molecular size
- ✓ Conjugation
- ✓ Conformation
- ✓ Oligomeric state

Maximize confidence

MALS eliminates the uncertainties of column calibration, molecular standards and non-ideal column interactions. Even composite molecules such as proteins and co-polymers are amenable to accurate analysis. MALS affords extra-high sensitivity to aggregates.



Molar Mass



Size



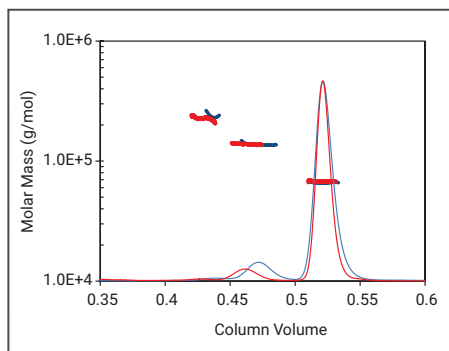
Conformation



Conjugation

Minimize consumption

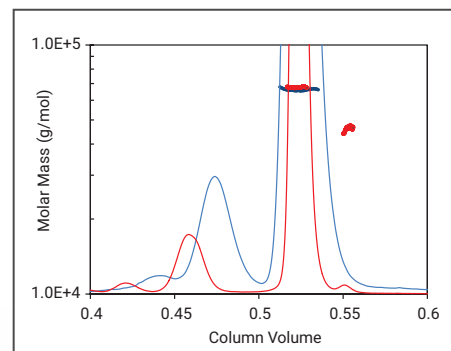
UHPLC affords many benefits such as dramatically reduced consumption of sample and solvent as well as improved separation. Standard light scattering instrumentation is not suitable for UHPLC's ultra-narrow peaks. The μDAWN and UT-rEX maintain the high resolution and low dispersion necessary for high-quality data with such small sample quantities.



Molar masses of a protein monomer, dimer, and trimer, determined by μSEC-MALS (red) and standard SEC-MALS (blue), showing identical results. Separation by 15 cm BEH-200 column (Waters).

Maximize productivity

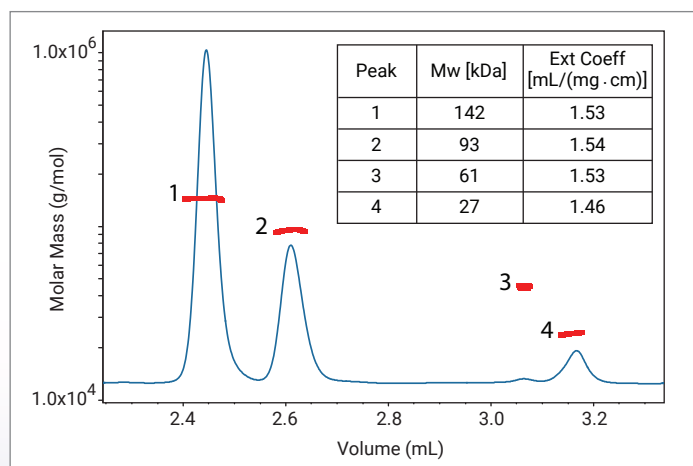
- Every μDAWN instrument includes a COMET™ ultrasonic cell cleaner for automated clean cycles.
- Add a WyattQELS™ dynamic light scattering module to measure sizes below 10 nm in radius.
- Combine standard and advanced SEC-MALS analyses with UHPLC's higher throughput.



UHPLC and HPLC light scattering chromatograms of the same protein analyzed to the left. Here UHPLC was run on a 30 cm BEH-200 column (Waters), revealing a fragment peak that was not discernable on HPLC or a 15 cm UHPLC column.

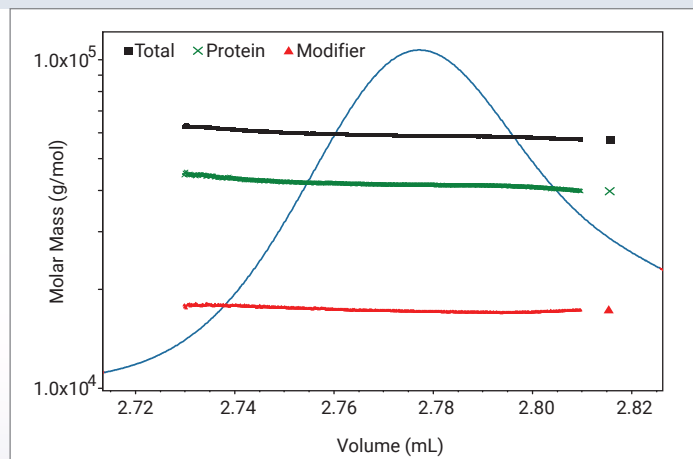


The μDAWN & UT-rEX interface with any UPLC® or UHPLC system.



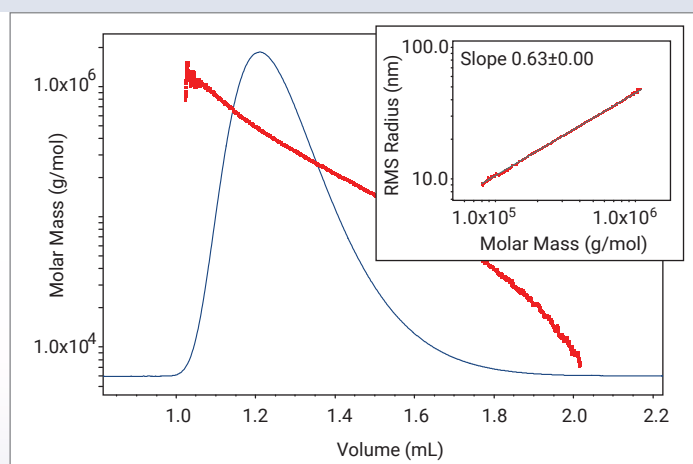
Identification of Protein Degradation Products

IgG and its fragments were identified and quantitated by μ SEC-MALS using a Waters Acquity UPLC system with a BEH-200 column, μ DAWN and Optilab UT-rEX. Mobile phase was phosphate-buffered saline at pH 7.4. Since the extinction coefficients are nominally unknown, molar masses of the protein fragments could only be determined by MALS with dRI concentration measurements. Simultaneously the extinction coefficients were determined by comparing UV and dRI peak areas. The combination of molecular weight and extinction coefficient helps confirm the identity of peaks 2, 3 and 4 as IgG fragments.



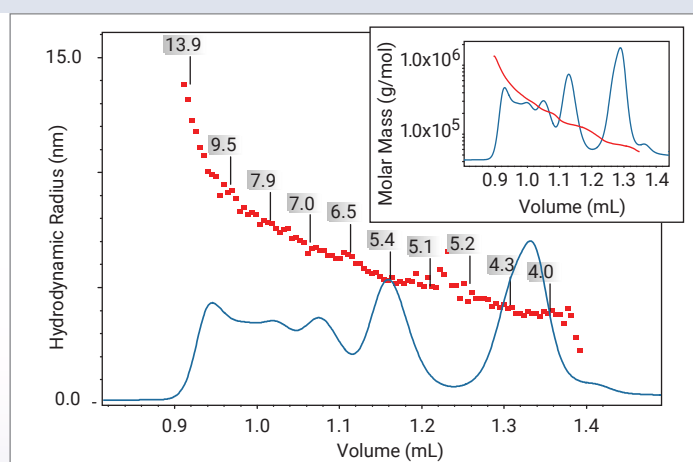
Protein Conjugate Analysis

The combination of UV, MALS and dRI may be used to analyze the composition of protein-carbohydrate conjugates, surfactant-enveloped membrane proteins and co-polymers. Here μ SEC-MALS, using a Waters Acquity UPLC system with a BEH-200 column, μ DAWN and Optilab UT-rEX, analyzed a glycoprotein to determine the molar mass of the protein and total glycan molecular weight, as well as the overall molecular weight of the complex.



Polymer Molecular Weight and Size

MALS provides first-principles determination of absolute molecular weight and size for the most reliable characterization of polymers. Here μ SEC-MALS, using a Waters Acquity UPLC system with APC XT 900 and XT 450 columns in series, μ DAWN and Optilab UT-rEX were used to analyze a polystyrene reference sample, NIST 706a. The heterogeneity of this sample is evident in the sloping molecular weight distribution across the peak. The inset shows a conformation plot obtained by mapping the root mean square radius vs. molar mass. The slope of the plot indicates that this polymer is in fact a linear, random coil.



Hydrodynamic Size of Proteins and Aggregates

While MALS cannot measure the size of molecules smaller than 10 nm in radius, with the addition of a WyattQELS embedded dynamic light scattering module, the μ DAWN can determine size (as well as molar mass) of smaller molecules. Here a stressed protein was analyzed to assess both hydrodynamic radius r_h and molecular weight Mw of the monomer, low oligomers and larger soluble aggregates. Sample was bovine serum albumin prepared in 10 M urea and subsequently dialyzed to phosphate-buffered saline, pH 7.4. Inset show molar mass distribution as determined by μ SEC-MALS. Dimer, trimer and tetramer peaks are clearly separated and identified by Mw.

Specifications

MALS Measurements

Molar Mass Range	200 Da – 10 MDa (proteins) or 1 MDa (linear polymers)*
Size Range (r_g)	10 nm – 50 nm, up to 150 nm with shape-specific models
Sensitivity	1.0 $\mu\text{g/mL}$ BSA, typical of 70 ng loading on 15 cm UHPLC-SEC column
Number of detector angles	3 light scattering detectors + forward monitor to correct for absorbing samples

DLS Size Measurement

(with WyattQELS module installed)

Size Range (Radius, r_h)	0.5 to 30 nm, flow rate 0.3 mL/min 0.5 to 20 nm, flow rate of 0.5 mL/min
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Sample Temperature Control

Ambient

Laser Properties

$\lambda = 658 \pm 4 \text{ nm}$; Laser power controllable 10% - 100% of maximum

Dispersion

(Increase in FWHM @ 0.3 mL/min)
Upstream detector to μDAWN : < 1.5 μL ;
Upstream detector to downstream detector: < 1.6 μL

Wetted Materials

316 stainless steel, fused silica, PEEK, PTFE, MP35N iron-free alloy

Electronics

Analog inputs	4 differential analog inputs with 24 bit resolution.
Additional Inputs/outputs	Alarm In/Out, Autoinject In/Retransmit, Recycle, 3 programmable analog outputs
Front Panel	162.5 mm high-resolution touch screen + physical keypad
Data Rate	Up to 73 Hz

Options

WyattQELS embedded dynamic light scattering module

Dimensions

60 cm (L) x 36 cm (W) x 17 cm (H)

* Depending on dn/dc , the sample concentration, and chromatography conditions, this is typical.
The upper size limit is set by the rms radius of the molecule, with a maximum size for reliable Mw of 50 nm.

Wyatt Technology is committed to continual improvement. Specifications subject to change without notice.

Warranty All Wyatt instruments are guaranteed against manufacturing defects for 1 year.



With installations in *more* than 65 countries, *more* than 11,000 refereed journal publications citing its instruments, and 18+ PhD scientists on site, Wyatt Technology is the **world's leading manufacturer of instruments** for absolute macromolecular characterization. It is the only company in the world focused exclusively on such systems, their design, and their applications.

DAWN, HELEOS, TREOS, Optilab, ViscoStar, NanoStar, Calypso, Möbius, MöbiuS, ASTRA, DynaPro, DYNAMICS, Aurora, International Light Scattering Colloquium, Light Scattering University, Light Scattering for the Masses, Protein Solutions, Wyatt Technology are registered trademarks of Wyatt Technology Corporation. UPLC is a registered trademark of Waters Corporation. Wyatt Technology instruments, components and software are covered by one or more of the following: U.S. Patent Nos.: 6,411,383; 6,426,794; 6,452,672; 6,519,032; 6,651,009; 6,774,994; 6,819,420; 6,975,392; 7,027,138; 7,283,221; 7,331,218; 7,386,427; 7,813,882; 7,911,594; 7,982,875; 8,195,405; 8,441,638; 8,525,991; British Patent Nos.: EP 0 710 831; EP 0 665 433; EP 1 134 577; EP 1 510 807; EP 1 517 143; EP 1 538 435; EP 1 507 136; EP 1 645 864; French Patent Nos.: EP 1 517 143; EP 1 645 864; German Patent Nos.: 694 30 918.4-08; 694 33 615.7-08; 601 31 486.7-08; 603 19 078.2-08; 60 2004 022 625.4-08; 60 2004 038 882.3; 60 2004 039 666.4; 60 2005 040 312.4; Japanese Patent Nos.: 4,439,211; 4,381,914; 4,426,951; 4,594,206; 4,680,402; 4,786,906; 4,813,784; 5,261,720; 5,500,365; Chinese Patent Nos.: ZL 2004 1 0070894.8; ZL 2004 1 0080545.4; ZL 2004 1 0070023.6; ZL 2004 1 0062673.6; ZL 2005 1 0108269.2; ZL 2011 1 0008100.5; Korean Patent No.: 794,478. Other patents pending. No part of this brochure may be reproduced in any way without written permission from Wyatt Technology Corporation.

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